Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **3 marks**.

1) Write a Python program to accept the marks of a student in **5 subjects**. Calculate the total marks, average, and percentage using arithmetic operators. Display the grade based on the following conditions:

- 90–100: Grade A
- 75–89: Grade B
- 60–74: Grade C
- 50–59: Grade D

```
# Accept marks of 5 subjects

m1 = float(input("Enter marks in Subject 1: "))

m2 = float(input("Enter marks in Subject 2: "))

m3 = float(input("Enter marks in Subject 3: "))

m4 = float(input("Enter marks in Subject 4: "))

m5 = float(input("Enter marks in Subject 5: "))

# Calculate total, average and percentage

total = m1 + m2 + m3 + m4 + m5

average = total / 5

percentage = (total / 500) * 100


# Display results

print("Total Marks =", total)

print("Average =", average)

print("Percentage =", percentage)
```

```
# Display grade using if-else

if percentage >= 90:

    print("Grade A")

else:

    if percentage >= 75:

        print("Grade B")

    else:

        if percentage >= 60:

            print("Grade C")

        else:

            if percentage >= 50:

                print("Grade D")

            else:

                print("Fail")
```

2) A student is eligible for a scholarship if:

- Marks are **80 or above**, and
- Family income is **less than ₹3,00,000**.
  Write a Python program to check the eligibility please give a python code for this

```
    marks = float(input("Enter the marks: "))

income = float(input("Enter the family income: "))

if marks >= 80 and income < 300000:

    print("Student is eligible for scholarship")

else:

    print("Student is not eligible for scholarship")
```

3) Write a Python program to simulate an ATM withdrawal. Accept the account balance and withdrawal amount. Check the following conditions:

- Withdrawal amount must be positive.
- Withdrawal amount must be a multiple of ₹100.
- Withdrawal amount must not exceed the available balance.

```
# Accept account balance and withdrawal amount

balance = int(input("Enter account balance: ₹"))

withdrawal = int(input("Enter withdrawal amount: ₹"))



# Check whether withdrawal amount is positive

if withdrawal > 0:


    # Check whether withdrawal amount is a multiple of 100

    if withdrawal % 100 == 0:


        # Check whether withdrawal amount is within balance

        if withdrawal <= balance:

            balance = balance - withdrawal

            print("Withdrawal successful")

            print("Withdrawn amount: ₹", withdrawal)

            print("Remaining balance: ₹", balance)

        else:

            print("Insufficient balance")


    else:
```

```
        print("Withdrawal amount must be a multiple of ₹100")


else:
    print("Withdrawal amount must be positive")
```

4) Write a Python program to accept a three-digit number and calculate:

- Sum of its digits
- Product of its digits
- Reverse of the number
- Whether the number is a palindrome
- Whether the sum of its digits is divisible by 3

```
# Accept a three-digit number

num = int(input("Enter a three-digit number: "))


# Check whether the number is a three-digit number

if num >= 100 and num <= 999:


    # Extract the digits

    digit1 = num // 100

    digit2 = (num // 10) % 10

    digit3 = num % 10


    # Calculate sum and product

    digit_sum = digit1 + digit2 + digit3

    digit_product = digit1 * digit2 * digit3
```

```
# Calculate reverse
reverse = digit3 * 100 + digit2 * 10 + digit1


# Display results
print("Sum of digits =", digit_sum)
print("Product of digits =", digit_product)
print("Reverse of number =", reverse)


# Check palindrome
if num == reverse:
    print("The number is a palindrome")
else:
    print("The number is not a palindrome")


# Check whether sum of digits is divisible by 3
if digit_sum % 3 == 0:
    print("The sum of digits is divisible by 3")
else:
    print("The sum of digits is not divisible by 3")


else:
    print("Please enter a three-digit number")
```

5) A customer purchases 5 items. Write a Python program to accept the price of each item using a loop and calculate the total bill.

```
total = 0

for i in range(1, 6):

    price = float(input("Enter the price of item " + str(i) + ": "))

    total = total + price

print("Total bill = ₹", total)
```

6) Write a Python program to accept the temperature of a city and display:

- **"Cold"** if temperature is below 20°C
- **"Moderate"** if temperature is between 20°C and 30°C
- **"Hot"** if temperature is above 30°C.

```
# Accept temperature

temperature = float(input("Enter the temperature in °C: "))



# Check temperature

if temperature < 20:

    print("Cold")

else:

    if temperature <= 30:

        print("Moderate")

    else:

        print("Hot")
```

7) Write a Python program to accept a person's age and calculate the movie ticket price:

- Below 12 years → ₹100
- 12–59 years → ₹200
- 60 years and above → ₹120

```
# Accept person's age

age = int(input("Enter your age: "))



# Calculate ticket price using if-else

if age < 12:

    ticket_price = 100

else:

    if age <= 59:

        ticket_price = 200

    else:

        ticket_price = 120

# Display ticket price

print("Movie Ticket Price = ₹", ticket_price)
```

8) Write a Python program to accept the expenses for 7 days using a loop and calculate the total weekly expense.

```
# Initialize total expense

total = 0

# Accept expenses for 7 days

for day in range(1, 8):

    expense = float(input("Enter expense for Day " + str(day) + ": "))

    total = total + expense

# Display total weekly expense

print("Total weekly expense = ₹", total)
```

9)Write a Python program to accept the amount of water consumed by a person for 7 days and calculate the average daily water consumption.

```
# Initialize total water consumption

total = 0


# Accept water consumption for 7 days

for day in range(1, 8):

    water = float(input("Enter water consumed on Day " + str(day) + " in litres: "))

    total = total + water

# Calculate average

average = total / 7

# Display results

print("Total water consumed =", total, "litres")

print("Average daily water consumption =", average, "litres")
```

10)A parking area charges:

- First 2 hours → ₹30
- 3–5 hours → ₹50
- More than 5 hours → ₹100

Write a Python program to accept the number of hours a vehicle is parked and calculate the parking fee.

```
# Accept number of parking hours

hours = int(input("Enter the number of hours parked: "))


# Calculate parking fee

if hours <= 2:
```

```
    fee = 30
else:
    if hours <= 5:
        fee = 50
    else:
        fee = 100
# Display parking fee
print("Parking Fee = ₹", fee)
```